

Graph-Based Reasoning for Abnormality Detection in Musculoskeletal X-Ray Images

Impact Statement

This work presents a lightweight graph-based deep learning framework for musculoskeletal abnormality detection in radiographic images. Unlike conventional convolution-based approaches that primarily focus on local visual features, the proposed method models spatial relationships between anatomical regions using graph attention mechanisms. The study demonstrates that graph-based reasoning can improve structural understanding in radiographic analysis while maintaining relatively low computational requirements.

The proposed framework may contribute to the development of efficient computer-aided diagnostic systems, particularly in settings where computational resources are limited. By combining convolutional feature extraction with graph-based relational learning, the work also provides a practical direction for incorporating structural context into medical image classification tasks.

Beyond musculoskeletal radiographs, the proposed approach may support future research involving graph representations in other medical imaging applications where spatial relationships between anatomical structures are important for diagnosis.